MobileViT-Based Traffic Analysis for Digital Twin Optimization

EPL 445 – Digital Video Processing | Project Proposal (Interim)

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1. Project overview

We will develop a web-based video analytics system for Topic 3 (Object Recognition in video). The system will detect and categorize vehicles in traffic footage, then aggregate detections into counts and density estimates that can be consumed by a city "digital twin" for traffic-flow monitoring and optimization.

2. Scientific goal and research question

Scientific goal: provide intelligent, automated interpretation of traffic video to support decision making.

Research question: How can we estimate traffic density and vehicle categories in near real-time using a lightweight transformer backbone (MobileViT)?

3. Data and datasets

We will use publicly available traffic video datasets with bounding-box labels (detection) and, when possible, tracking annotations. Candidate datasets include UA-DETRAC, BDD100K (video subset), and AI City Challenge. For evaluating the optional tracking module, we may also use tracking benchmarks such as VOT.

4. Methodology and system design

- Video ingestion: load files or streams (RTSP) and sample frames at a configurable rate.
- Pre-processing: resize/normalize frames; define Regions of Interest (ROIs) per lane or camera view.
- Model: MobileViT backbone + lightweight detection head to output vehicle bounding boxes and classes.

- Post-processing: non-max suppression; optional tracking (e.g., SORT/ByteTrack-lite) to stabilize IDs across frames.
- Aggregation: per-ROI counts and density estimates; time-series summaries (e.g., per minute).
- Web application: Python backend (e.g., FastAPI) exposing REST endpoints + simple dashboard.

5. Resource requirements and feasibility

- Software: Python, OpenCV, PyTorch, torchvision; optional PyTorchVideo; scikit-learn for metrics.
- Models: start from pre-trained weights to enable fast “first runs”; fine-tune on traffic data if time permits.
- Hardware: GPU recommended for near real-time inference; CPU-only mode retained for portability and basic demos.
- Feasibility plan: deliver an end-to-end working pipeline early, then iterate on accuracy and speed.

6. Evaluation plan and metrics

We will evaluate on a dedicated evaluation/testing set (and optionally training set for reference).

- Primary metrics (requested in the course brief): Accuracy, Sensitivity (Recall), Specificity, F-Score; ROC curve and AUC where applicable.
- Detection metrics: Precision/Recall and mean Average Precision (mAP).
- Counting metrics: Mean Absolute Error (MAE) / Mean Absolute Percentage Error (MAPE) vs. ground truth counts.
- Performance: frames-per-second (FPS) and end-to-end latency under fixed input resolution and hardware.

7. Milestones (Spring 2026)

- 06 Apr 2026 – Interim Presentation #1: introduction, methodology, required resources, feasibility plan.
- By 23 Apr 2026 – Interim Presentation #2 (“first runs”): run the light version of the pipeline and report metrics on the evaluation/testing set.
- Early May 2026 – Improve robustness: ROI tuning, tracking integration, speed/accuracy trade-offs.
- 25 May 2026 – Final presentation & live demo (web-based application) + final report submission.

8. Risks and mitigations

- Dataset mismatch (camera angle / labels): start with one dataset and lock label mapping early; add a second dataset only after baseline works.
- Real-time performance constraints: benchmark early; add frame skipping/resolution scaling; keep a CPU fallback for demo.
- Detection-to-counting errors (double counts): use ROI rules + tracking; validate counts against short manually checked clips.

9. References (starter)

• Course brief: EPL 445 Final Project – 2026 (University of Cyprus, 19 March announcement).

• MobileViT paper and model documentation (to be added).

• Selected dataset papers/pages (UA-DETRAC, BDD100K, AI City Challenge) (to be added).